Speech Notes for the Conservative Formula

Yingni Chang and Yikai Shen

Read-aloud script: approximately 2,635 words, or 20 minutes at 130 words per minute. Allow 20-23 minutes with pauses. Cues are reminders, not spoken lines. Optional split: Yingni, slides 1-9; Yikai, slides 10-21.

Slide 1 | Replicating the Conservative Formula

Cue: Introduce both authors and the purpose of the project.

Hello everyone. We're Yingni Chang and Yikai Shen. Our project implements the Conservative Formula proposed by David Blitz and Pim van Vliet. The idea is to combine low volatility, momentum and shareholder payout in a portfolio of one hundred stocks.

We first try to reproduce the US results over 1929 to 2016, which is the paper's sample. We then extend the backtest through 2024. The historical comparison is fairly close, but the recent results are weaker after costs and factor adjustment. I'll explain the rule first, then the data and results, including the payout and sector issues raised in the assignment.

Transition: Let me start with what we wanted to test.

Slide 2 | What we test

Cue: Follow the three boxes from left to right: replicate, explain, then test recent performance.

The three boxes show our research questions from left to right. First, can we reproduce the paper's return, volatility and turnover? Second, how much of the return is explained by familiar equity factors? Third, what happens when we move to recent periods and include trading costs?

These questions matter because a strong result over almost ninety years does not necessarily mean that the strategy still has a useful advantage today. We use the Carhart model as our main factor adjustment and compare gross returns with returns at ten and thirty basis points per dollar traded. The fixed comparison periods were chosen before computing the post-2016 portfolio results.

Transition: Those comparisons depend on the data, so I'll turn to the CRSP inputs.

Slide 3 | The data

Cue: Point to the coverage table, then the three cleaning choices on the right.

We use the full monthly CRSP files from WRDS. After cleaning, the panel has about 3.95 million security-month observations and 26,746 securities. We combine stock returns with historical company information, delistings, distributions and the market index. Fama-French factor returns help evaluate performance after the portfolio is constructed.

Three data-cleaning choices are important. First, we keep stocks that disappear, including their terminal returns, instead of looking only at survivors. Second, we use one common share class per company each month to avoid double counting. Third, we respect the smaller early universe: only 393 stocks are eligible at the first formation. The universe first reaches one thousand in June 1964, so we use the available eligible stocks in the early backtest.

Transition: With those files cleaned, the selection rule is fairly simple.

Slide 4 | The portfolio rule

Cue: Trace the Conservative flow, compare the two columns, then distinguish CMS from either long-only portfolio.

The left flowchart builds the Conservative portfolio. We first require thirty-six consecutive months of return history, then choose the largest one thousand eligible US stocks and keep the lower-volatility half. The volatility formula annualizes the standard deviation of thirty-six monthly total returns using the square root of twelve.

The momentum formula compounds eleven monthly price returns and skips the latest month. We average the momentum and net payout ranks to select the best one hundred. The right-hand table shows the Speculative mirror: the higher-volatility half and the weakest combined momentum and net payout scores. Both portfolios hold one hundred stocks at equal initial weights and rebalance quarterly; weights drift between rebalances.

Each portfolio on its own is long-only. Conservative-minus-Speculative, or CMS, instead buys Conservative and shorts Speculative. It measures the contrast between the groups and differs from holding Conservative alone. The early sample has fewer eligible stocks, so the full one-thousand-to-five-hundred counts do not apply throughout. The history screen precedes the size cut, and holdings earn returns after formation.

Transition: I'll show how we calculate the three inputs to those ranks.

Slide 5 | Building the signals

Cue: Explain each signal briefly; spend most of the time on payout.

Volatility is the standard deviation of thirty-six monthly total returns, expressed in annual units. Momentum compounds eleven monthly price returns and skips the most recent month. Portfolio performance, in contrast, uses total returns including distributions and delistings.

The formula at the top separates net payout into two terms. The first fraction is twelve months of adjusted cash dividends divided by the current adjusted price. The second term is one minus current adjusted shares divided by their twenty-four-month average. Both windows include the formation month. A falling share count raises the signal; issuance lowers it. This is a proxy for net repurchases, not observed buyback cash spending. Prices and shares are split-adjusted so a stock split does not create an artificial buyback. The twelve-month dividend window is a disclosed convention because the paper does not fully specify that part.

Transition: Now we can compare the backtest with the published figures.

Slide 6 | Our results versus the paper

Cue: Point to the return/risk comparison and the growth chart.

Over the paper's sample, our Conservative portfolio compounds at 14.80 percent a year before costs, compared with 15.10 percent in the paper. Our volatility is 17.06 percent, compared with 16.50 percent. The broad return-and-risk result is therefore close.

The main unresolved difference is quarterly turnover: we get 38.9 percent, compared with 32.0 percent. Some control and regression results differ as well. Share-class choices, capitalization and dividend estimation could contribute, but we cannot separate their effects. We leave these gaps visible rather than adjusting the strategy to force a match.

Transition: Close raw returns are only part of the comparison; let's look at factor-adjusted returns.

Slide 7 | Factor-adjusted returns

Cue: Define alpha before discussing the factor models.

Alpha here is the return left after accounting for the factors in the regression. A high raw return can still have little alpha if the portfolio mainly holds known factor exposures.

For the Conservative long-only portfolio, annual Carhart alpha over the paper sample is 3.23 percent. When we use the five Fama-French factors plus momentum over 1963 to 2016, the estimate falls to -0.11 percent. Adding profitability and investment changes the interpretation substantially. We also compare the Conservative-minus-Speculative contrast with the paper. The post-1963 comparisons are closer than the full-sample ones, but that does not establish the cause of the remaining difference.

Transition: Trading costs are another reason gross return can be misleading.

Slide 8 | How costs change the result

Cue: Explain basis points and the single-counted turnover convention.

We charge costs on both purchases and sales. Thirty basis points means a charge of zero point three percent per dollar traded. Quarterly turnover on the slide is single-counted, so a purchase and sale are not counted twice in that measure.

At thirty basis points, the paper-period annual compound return falls to 13.73 percent. For 1990 to 2024, the break-even cost for Carhart alpha is approximately 36 basis points. Break-even means the cost at which the estimated advantage reaches zero. In the period after 2016, gross excess return and alpha are already nonpositive. These are flat cost scenarios, not a complete model of historical liquidity or market impact.

Transition: Let's look at how the estimates change across periods.

Slide 9 | Results by period

Cue: Teal solid = gross; copper dashed = net 30 bps. Trace the early-to-recent pattern.

The subperiods show why the long history needs to be broken up. The earlier Carhart estimates are stronger, while the recent estimates are weaker. The rolling plots show a similar pattern, although adjacent windows overlap and should not be treated as independent observations.

For 2017 to 2024, gross Carhart alpha is -1.21 percent. Its ninety-five-percent interval runs from -5.12 percent to 2.69 percent. That interval is wide because the sample contains only eight years. We do not find strong evidence of outperformance in that period, but an insignificant estimate is not proof that expected alpha is exactly zero.

Transition: We will now focus on the recent sample in more detail.

Slide 10 | Trading costs: two US samples

Cue: Compare the two periods at the same cost level.

The left table covers 1990 to 2024, our broader recent sample. The right table isolates 2017 to 2024, which is beyond the paper's original period and addresses the professor's suggested extension. Both tables test zero to thirty basis points in steps of five. Within each period, the holdings and trades are identical across cost assumptions, and both purchases and sales pay costs.

For 1990 to 2024, CAGR falls from 12.24 percent before costs to 11.17 percent at thirty basis points, against market CAGR of 10.10 percent. Annual Carhart alpha falls from 1.17 percent to 0.20 percent. In 2017 to 2024, CAGR falls from 11.01 percent to 9.97 percent, while the market compounds at 13.01 percent. The strategy makes money in absolute terms but trails the market even before costs. Annual alpha falls from minus 1.21 percent to minus 2.10 percent.

CAGR describes compound growth. Above-market return is the annual arithmetic strategy-minus-market difference, while alpha controls for Carhart factors. All fourteen alpha confidence intervals include zero. The periods overlap, so this is a comparison of sample definitions, not two independent experiments. Lower costs improve returns, but do not establish positive alpha in the extension.

Transition: Next are the two implementation checks highlighted in the assignment.

Slide 11 | Dividends and share repurchases

Cue: Emphasize the assignment's payout warning and the matched comparison.

This is one of the specific issues raised in the assignment. Dividend yield alone misses the indirect payout from repurchases. For example, with a three-percent dividend yield, ninety adjusted shares today and a one-hundred-share average, the net payout signal is thirteen percent. Issuance works in the opposite direction.

In the matched backtest over the paper period, the net-payout rule earns an arithmetic 15.33 percent a year before costs, against 14.11 percent using dividends alone. The share-count term adds 1.23 percent a year, and 1.63 percent over 1990 to 2024. We hold the other rules constant so that we are comparing the payout ranking. This shows that the share-count term matters. It is still a proxy for net repurchases after issuance, not directly observed buyback cash spending or a causal estimate of the effect of buybacks.

Transition: The other concern is industry positioning.

Slide 12 | Industry allocation and stock selection

Cue: Explain the reference universe and then the industry quota rule.

For sector positioning, we use Fama-French twelve industries. We compare our portfolio with a cap-weighted reference from the same formation universe, so a broad-market universe difference does not get mistaken for industry allocation.

Over the paper sample, industry allocation contributes about 0.82 percent a year and stock selection about 3.80 percent. A separate universe-to-market term completes the monthly identity. These are arithmetic contributions, not contributions to CAGR.

We also follow the course's quota idea: if an industry contains fifteen percent of universe stocks, it receives about fifteen of the one hundred holdings. That variant earns an arithmetic 15.18 percent a year before costs, 0.15 percentage points below the baseline. It changes the paper's global screen and was considered after the baseline results, so we treat it as an exploratory comparison.

Transition: We also test the signals outside the original stock-selection rule.

Slide 13 | Signal tests

Cue: Define IC and explain why rank correlation differs from a mean return spread.

We next examine the signals separately. We remove industry means, standardize the exposures and clip extremes. The information coefficient, or IC, is the rank correlation between a signal and later industry-adjusted returns.

The composite's one-month IC is about 0.047. We also look at deciles, persistence and longer holding horizons. A positive rank correlation does not always imply a strong mean spread when returns are skewed. The factor-mimicking portfolios help separate signal exposures from industry, size and historical Fama-French controls. Their gross information ratio falls as we average more old portfolios, along with turnover. They are diagnostic portfolios with leverage, rather than a fully costed trading strategy.

Transition: That leads to the trading and portfolio-construction experiments.

Slide 14 | Long/short and linear optimization

Cue: Separate the failed trading experiment from the optimizer grid.

The long/short exercise buys the top composite decile and sells the bottom decile. We compare holding periods, trading costs and monthly borrow fees. The original buy-and-hold cohorts lose all capital in the early data for several holding periods. We then examine monthly reweighting, which still has an extremely deep drawdown. That alternative was chosen after seeing the losses.

The linear optimizer gives stronger-looking recent results for some settings. It caps individual stocks, constrains industries and limits turnover. But two of the four 95% intervals exclude zero for the four displayed recent examples, and those intervals are unadjusted for testing many settings. The post-2016 estimates are weaker. We would need an unseen test before treating the best grid result as reliable.

Transition: The next model adds a covariance penalty.

Slide 15 | Mean-variance portfolios

Cue: Use the table to compare the two risk-aversion settings.

The quadratic model adds portfolio variance to the objective, alongside expected return and trading cost. We estimate covariance using the last thirty-six monthly returns. With about one thousand stocks and only thirty-six observations, sample covariance needs regularization, so we combine it equally with diagonal covariance.

For 1990 to 2024, risk aversion five gives net CAGR of 11.59 percent and volatility of 12.51 percent. At risk aversion twenty, those figures are 10.39 percent and 11.51 percent. More risk aversion lowers realized volatility, but market correlations remain high. These long-only portfolios are not market-neutral. We chose these settings after examining the baseline and have not established that they are optimal.

Transition: Finally, we test a simpler change aimed at reducing turnover.

Slide 16 | A buffer to reduce trading

Cue: Point to the turnover reduction and the confidence interval.

The buffer asks whether we need to sell a holding immediately when its rank slips below one hundred. We keep it while it remains eligible and among the best one hundred and fifty ranks, then fill vacancies. We fixed that buffer size before testing the variant.

Quarterly turnover falls from 39.8 percent to 28.4 percent. The net paired difference is 0.34 percent a year, but its interval includes zero. The modification reduces trading, while also changing holdings, so we cannot attribute the whole return difference to costs. We view it as a useful trading experiment, not proof of a better signal.

Transition: We now turn to the time-series and international extensions.

Slide 17 | When the recent return gap developed

Cue: Explain the comparison and its limits.

The time-series view explains what an eight-year average can hide. The top panel shows the actual wealth paths. The lower panel compares the most recent twelve months of strategy and market returns. The sign can change during the sample. We also calculate trailing thirty-six-month Carhart alpha with Newey-West intervals in the notebook. Those windows overlap, and the short windows produce uncertain estimates. We use them to describe changing performance, rather than claim that we can forecast the next regime.

Slide 18 | Costs that change with quoted spreads

Cue: Explain the comparison and its limits.

The transcript also suggests scaling costs with average spreads. We use CRSP closing bid and ask quotes for the eligible large-stock universe. We fix the reference average using 2016, then use only the previous month's spread to set the next month's cost. A five-basis-point reference therefore need not remain five in every month. The chart shows the changing cost assumption and quote coverage. It is a market-wide quote proxy. It does not measure our actual execution price, commissions or market impact.

Slide 19 | UK data and implementation

Cue: Explain the comparison and its limits.

For the UK comparison, we download Compustat Global daily data and take each month's final available observation. We use ordinary London shares quoted in pounds, retain inactive companies, and use earlier observations to build the same thirty-six-month history. We adjust prices and shares for splits, then use the total-return factor to include distributions. Our benchmark holds the same eligible UK universe at capitalization weights. Current country and exchange headers and incomplete terminal payments make this a qualified extension. We do not use US factors to claim UK alpha.

Slide 20 | UK and US after the paper period

Cue: Explain the comparison and its limits.

At ten basis points, the US strategy returns 10.66% annually against 13.01% for its benchmark. The UK base sample returns 7.23% against 5.40%. The UK exit stress returns 1.65% against 3.55% for its comparator, reversing the relative advantage. Adding a fifty-basis-point purchase tax to the ordinary ten-basis-point costs gives 6.34% for the UK base case. The relative UK result is more encouraging in this sample, but pounds and dollars represent different investor exposures. The benchmark definitions also differ. We should check historical classifications, terminal payments and a common-currency benchmark before calling this a robust country effect.

Transition: Together, these results lead to our final conclusions.

Slide 21 | What we conclude

Cue: Close by connecting the data work, historical replication and both extensions.

We tested a simple rule using low volatility, momentum and net payout. The historical CRSP data need careful treatment of splits, delistings and signal timing. We reproduce much of the paper's historical return and risk pattern, but turnover and some factor-regression results still differ, so this is not an exact replication.

We address both assignment warnings: payout includes the split-adjusted share-count term, and industry positioning is measured and tested separately. Dividends alone lose useful information. The buffer reduces turnover, while the optimization experiments do not establish a stronger strategy on unseen data.

The combined cost table distinguishes the broader 1990 to 2024 sample from the 2017 to 2024 extension. The extension earns a positive absolute return but trails the market even before costs. Lower costs improve returns without establishing positive alpha. Rolling windows show that performance changes through time; spread-scaled costs are a liquidity sensitivity rather than actual execution costs.

The UK base case performs better relative to its own benchmark, but a severe exit-loss stress reverses that advantage. Purchase taxes, different currencies, benchmark definitions and incomplete terminal payments limit the comparison. Overall, the formula is understandable and useful for studying factor investing. Our evidence does not establish persistent outperformance. Thank you.

Questions to Prepare For

Did you reproduce the paper exactly?

No. Return and volatility are close, but quarterly turnover is 38.9% versus 32%, and some regression/control results differ. We cannot isolate the cause without the authors' exact cleaned inputs and conventions.

How do you account for share repurchases?

Payout is dividend yield plus one minus latest split-adjusted shares divided by their 24-month average. Repurchases raise the signal; issuance lowers it. We check split neutrality and compare the rule with dividends alone. Share changes are a net proxy, not reported buyback cash flow.

Why isn't the paper baseline sector-neutral?

The paper explicitly compares scores across sectors. We preserve that rule for the replication and separately use industry-adjusted diagnostics, allocation/selection attribution and a count-proportional FF12 quota strategy.

Does an insignificant alpha mean the strategy does not work?

It means the estimate does not provide strong evidence of nonzero factor-adjusted return at the chosen confidence level. It does not prove expected alpha is exactly zero. The strategy can still provide lower volatility and exposure to known factors.

Which experiments use an unseen sample?

The main comparison periods were fixed before computing post-2016 portfolio returns. The later sector and quadratic experiments, and the monthly-reweighted long/short, were considered after results were seen. We do not claim an unseen validation sample for those variants.

Are the capacity and cost assumptions realistic?

They are simplified scenarios. Costs are proportional to traded dollars, sometimes scaled by size and volatility. Capacity uses monthly volume and a 10% participation assumption, normalized to 2024 market size. The follow-up also uses lagged quoted spreads and a separate UK purchase-tax scenario. Market impact, daily scheduling and actual borrow availability are omitted.